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Assignment Report

Term Report: Strategic Logistics Optimization via Mixed-Integer Linear Programming (MILP)

Subject: Comparative Analysis of Linear vs. Mixed-Integer Modeling in Supply Chain Networks

1. Executive Summary

This report evaluates the optimization of a distribution network consisting of two primary warehouses (A and B) and three demand centers (Cafes X, Y, and Z). The investigation focuses on the mathematical necessity of **Mixed-Integer Linear Programming (MILP)** when dealing with non-linear cost structures, specifically fixed "route-opening" fees.

Traditional Linear Programming (LP) often fails in industrial contexts where binary decisions, such as whether to activate a shipping lane or open a facility; incur costs independent of volume. In this scenario, a fixed fee of **\$70** on the Warehouse A to Cafe Z route serves as the primary optimizer constraint. While basic LP models might erroneously favor this route due to its variable cost efficiency, the MILP approach correctly identifies the global optimum by penalizing the binary activation of that route.

Through the application of the **"Big M" method** and a two-phase sensitivity analysis, this study yielded a **5.07% reduction** in total landed costs, dropping the total from a manually adjusted \$690.00 to an optimized **\$655.00**. Furthermore, the dual-phase solution identified that Warehouse B is the critical infrastructure bottleneck, with a shadow price of **-\$3.00**, indicating that marginal capacity expansion here is the most effective lever for future cost reduction.

2. Problem Formulation and Data Setup

The network is defined by the following supply and demand parameters:

- **Warehouse A Capacity:** 100 units
- **Warehouse B Capacity:** 125 units
- **Cafe X Demand:** 80 units
- **Cafe Y Demand:** 60 units
- **Cafe Z Demand:** 70 units

3. Methodology

3.1 Initial LP Implementation: The Naive Approach

The baseline model was constructed as a standard Transportation Problem using the PuLP library in Python. In this initial phase, the objective function was defined strictly by variable costs:

$$\text{Minimize } Z = \sum_w \sum_c (\text{variablecost}_{w,c} \cdot x_{w,c})$$

- **Variables:** Decision variables (x) were defined as continuous and non-negative.
- **Limitation:** The solver was "blind" to the fixed \$70 cost associated with route A to Z.
- **Result:** The model optimized for a variable cost of \$620.00, but once the \$70 fee was manually added post-optimization (because the route was used), the actual landed cost rose to **\$690.00**.

3.2 Phase 1: The MILP (Decision Logic & Big M)

To solve the problem holistically, the model was upgraded to a Mixed-Integer Linear Program.

- **Binary Variables (y):** A set of binary variables was introduced where $y_{w,c} = 1$ if a route is used, and 0 otherwise.
- **The Big M Constraint:** To link the continuous flow (x) with the binary switch (y), I implemented:

$$x_{w,c} \leq M \cdot y_{w,c}$$

Here, M (set to 1000) acts as an upper bound.

- **Integrated Objective:** The new objective function combined variable and fixed costs:

$$\text{Minimize } Z = \sum (x_{w,c} \cdot \text{varcost}_{w,c}) + \sum (y_{w,c} \cdot \text{fixedcost}_{w,c})$$

3.3 Phase 2: Sensitivity Analysis (The Fix and Re-solve Technique)

Since Shadow Prices (dual values) do not exist for MILP models due to the non-continuous feasible region, a two-step "Relaxation" strategy was employed:

1. **Freezing Decisions:** The optimal binary values (y) from Phase 1 were captured and converted into hard constraints (e.g., $x_{A,Z} = 0$).
2. **Continuous Re-solve:** A new LP was initialized using only continuous variables but restricted to the optimal architecture chosen by the MILP.
3. **Dual Extraction:** The standard LP solver was then able to access the *.pi* attribute of the supply constraints to find the Shadow Prices.

4. Results and Analysis

The transition from a basic Linear Program to a Mixed-Integer Linear Program (MILP) resulted in a fundamental shift in the network's logistics strategy. While the initial LP focused on local variable cost efficiency, the MILP optimized for the global minimum by recognizing the step-cost hurdle of route activation.

4.1 Optimal Shipping Configuration and Flow

The MILP model's primary strategic shift was the total abandonment of the Warehouse A to Cafe Z route. Despite having a variable cost of \$5, the \$70 fixed "gate fee" made it economically non-viable compared to routing Cafe Z's 70-unit demand through Warehouse B.

The Final Optimized Schedule:

- **Warehouse A Distribution:** * Ships **80.0 units** to Cafe X at \$2/unit.
 - Ships **5.0 units** to Cafe Y at \$4/unit.
 - **Utilization:** 85/100 units (15 units of slack capacity).
- **Warehouse B Distribution:**
 - Ships **55.0 units** to Cafe Y at \$1/unit.
 - Ships **70.0 units** to Cafe Z at \$6/unit.
 - **Utilization:** 125/125 units (Fully utilized).

4.2 Comparative Performance Metrics

A comparison between the naive LP and the MILP highlights the "hidden" costs associated with poor model selection.

Metric	Initial LP (Naive)	MILP (Optimized)	Delta
Variable Shipping Cost	\$620.00	\$655.00	+\$35.00
Fixed Route Fees	\$70.00	\$0.00	-\$70.00
Total Landed Cost	\$690.00	\$655.00	-\$35.00

The paradox of the result is that the "optimized" solution actually has a higher variable cost (+\$35) but achieves a lower total landed cost (-\$35) by avoiding the route activation fee entirely.

4.3 Efficiency and Slack Analysis

The analysis reveals a mismatch between capacity and geographic efficiency:

- **Efficiency Gap:** Warehouse A is under-utilized because its most direct route to the high-demand Cafe Z is blocked by an prohibitive fixed fee.
- **Operational Strain:** Warehouse B is operating at 100% capacity because it must cover the volume that Warehouse A cannot economically reach.

5. Sensitivity Insights (Dual Analysis)

By "fixing" the binary decisions and re-solving as a continuous LP, we extracted the shadow prices which reveal the true economic value of each warehouse's capacity.

- **Warehouse A (Shadow Price = \$0.00):** Since Warehouse A has **15.0 units of slack**, the marginal value of an additional unit of capacity is zero. There is no scenario in the current configuration where more capacity at A would lower the total cost.
- **Warehouse B (Shadow Price = -\$3.00):** Warehouse B is the system's "binding constraint". The shadow price of -\$3.00 indicates that if Warehouse B had one extra unit of capacity, the total system cost would drop by exactly \$3.00.

Marginal Savings Logic: Currently, Warehouse B is full, forcing the remaining 5 units of Cafe Y's demand to be fulfilled by Warehouse A at \$4/unit. If Warehouse B gained 1 unit of capacity:

1. B would ship 1 more unit to Y (Cost: +\$1).
2. A would ship 1 fewer unit to Y (Savings: -\$4).
3. **Net Change:** -\$3.00.

6. Conclusion

The implementation of a Mixed-Integer framework is essential for supply chain models where fixed costs are present. This methodology demonstrates that the "Fix and Re-solve" approach is a robust way to maintain the rigorous decision-making of MILP while regaining the analytical depth of LP sensitivity analysis.